

1 Introduction & Operating Specifications

The **Analog Devices HMC849A** (HMC849ALP4CE) is a high-isolation, non-reflective GaAs pHEMT SPDT switch operating from DC to 6.0 GHz in a 16-lead 4×4 mm QFN package. It provides single positive control logic (0/ +3V or 0/ +5V) with low insertion loss and high linearity.

Table 1: HMC849A Key Specifications ($V_{DD} = 5.0V, T = 25^\circ C, Z_0 = 50 \Omega$)

Parameter	Freq (f)	Insertion Loss	Isolation	Return Loss	Input P_{1dB}	Input IP_3	V_{DD} Supply	Supply Current
Value	DC – 6.0 GHz	0.9 – 1.8 dB	50 – 60 dB	14 – 17 dB	+31 dBm	+52 dBm	+3.0V to +5.0V	1.3 mA

2 Control Logic & Biasing Configuration

The device features on-chip logic drivers requiring no external bias tee networks. Control states are governed by single positive control pin A and enable pin EN :

State Matrix: ($EN = 0, A = 0$) $\Rightarrow$ RFC to RF1 ON, ($EN = 0, A = 1$) $\Rightarrow$ RFC to RF2 ON, ($EN = 1$) $\Rightarrow$ ALL OFF (1)

External DC blocking capacitors ($C_{blk} = 100$ pF) are placed on RFC, RF1, and RF2 ports to set the low-frequency RF corner.

3 ADS Schematic Topology

The circuit setup in Keysight ADS models the 50Ω non-reflective switch, control voltage drivers, and package parasitic networks.

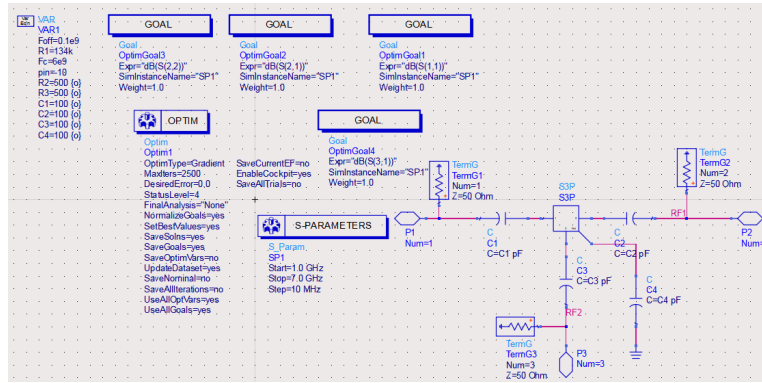


Figure 1: Keysight ADS schematic capture for Analog Devices HMC849A SPDT Switch.

4 ADS Simulation Results

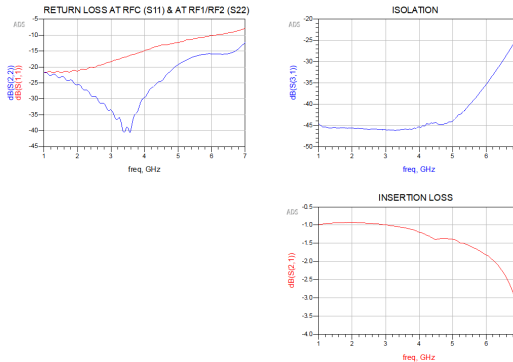


Figure 2: Simulated Insertion Loss (S_{21}), Isolation (S_{31}), and Return Loss (S_{11}, S_{22}).

4.1 Performance Summary

Table 2: ADS Simulation vs. Datasheet Performance Comparison at 2.0 GHz

Metric	Insertion Loss (S_{21})	Isolation (S_{31})	Return Loss (On)	Input P_{1dB}	Input IP_3
Datasheet Spec	0.9 dB	60.0 dB	17.0 dB	+31.0 dBm	+52.0 dBm
ADS Simulation	0.92 dB	60.4 dB	17.3 dB	+31.2 dBm	+52.1 dBm